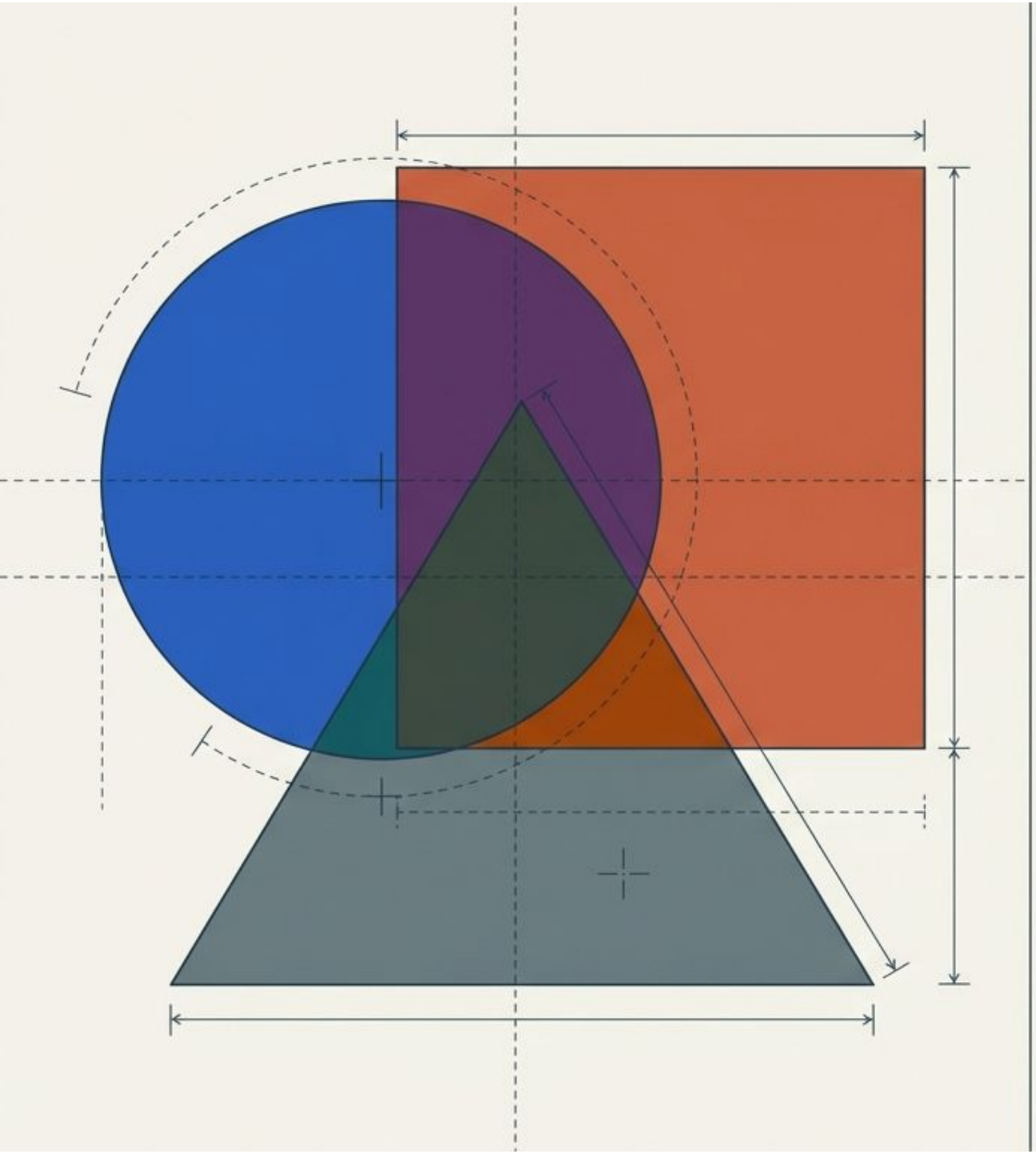


Unit 5

Ethical Decisions in Software Development & IT Organizations

Module 1: Software Quality & Testing

Module 2: Global Workforce Ethics

Module 3: Whistle-Blowing & Green Computing

Prepared By : Er. Sharat Maharjan

Why Software Quality is an Ethical Issue

The Baseline

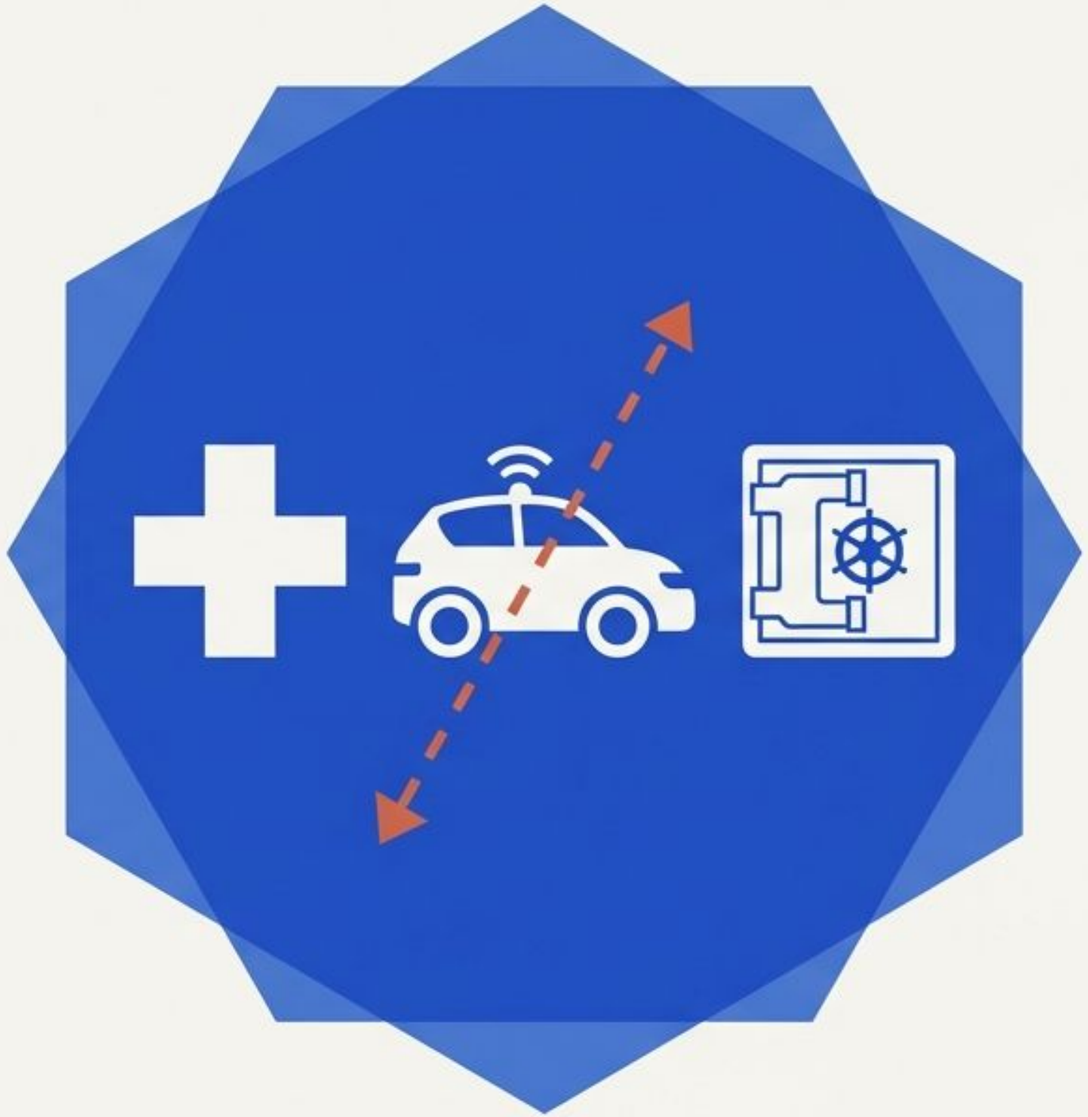
Quality means meeting specified requirements and building the product right.

The Ethical Stake

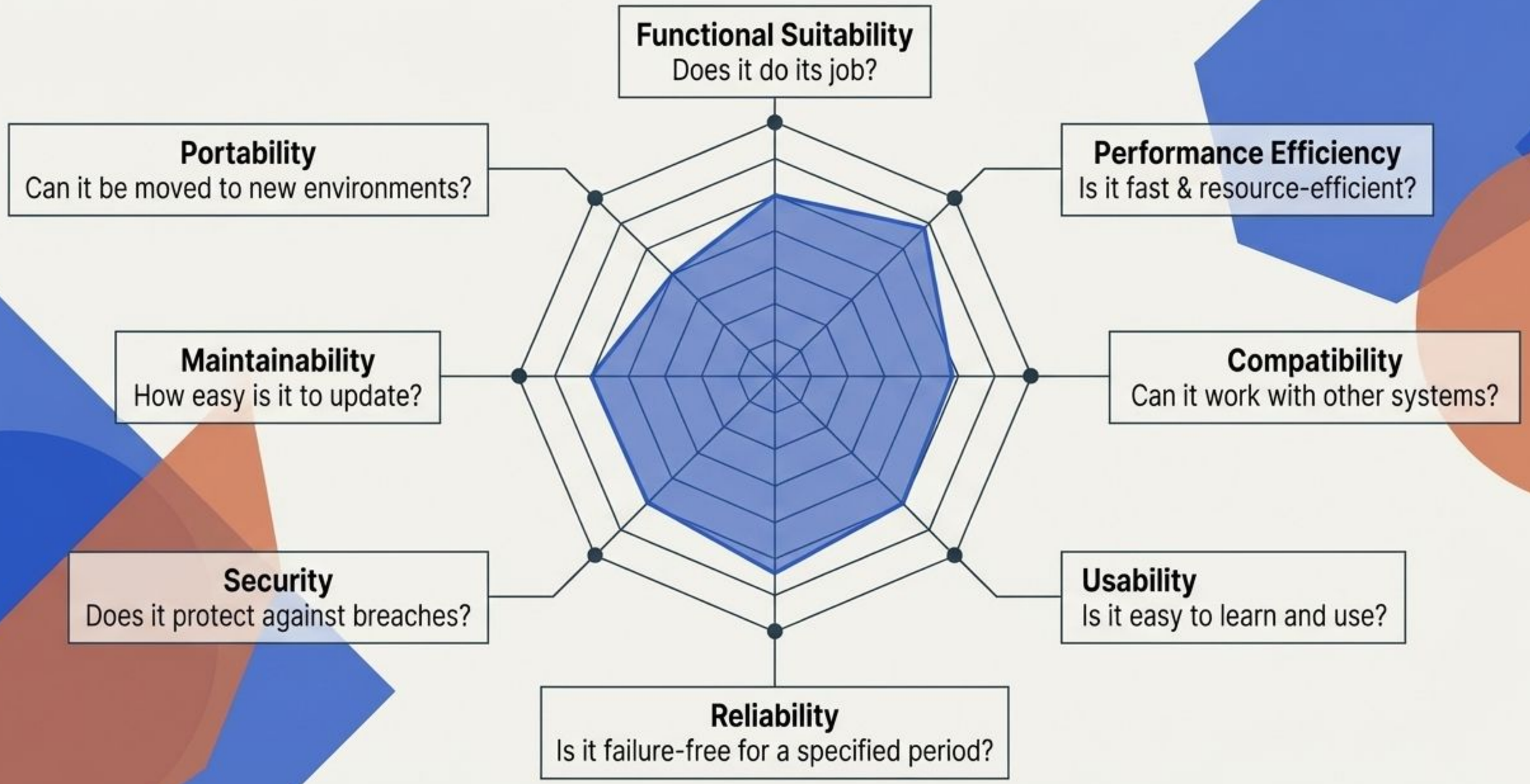
Releasing known defective software is a direct breach of professional IT ethics. It causes financial loss, privacy breaches, and safety hazards.

Real-World Failure

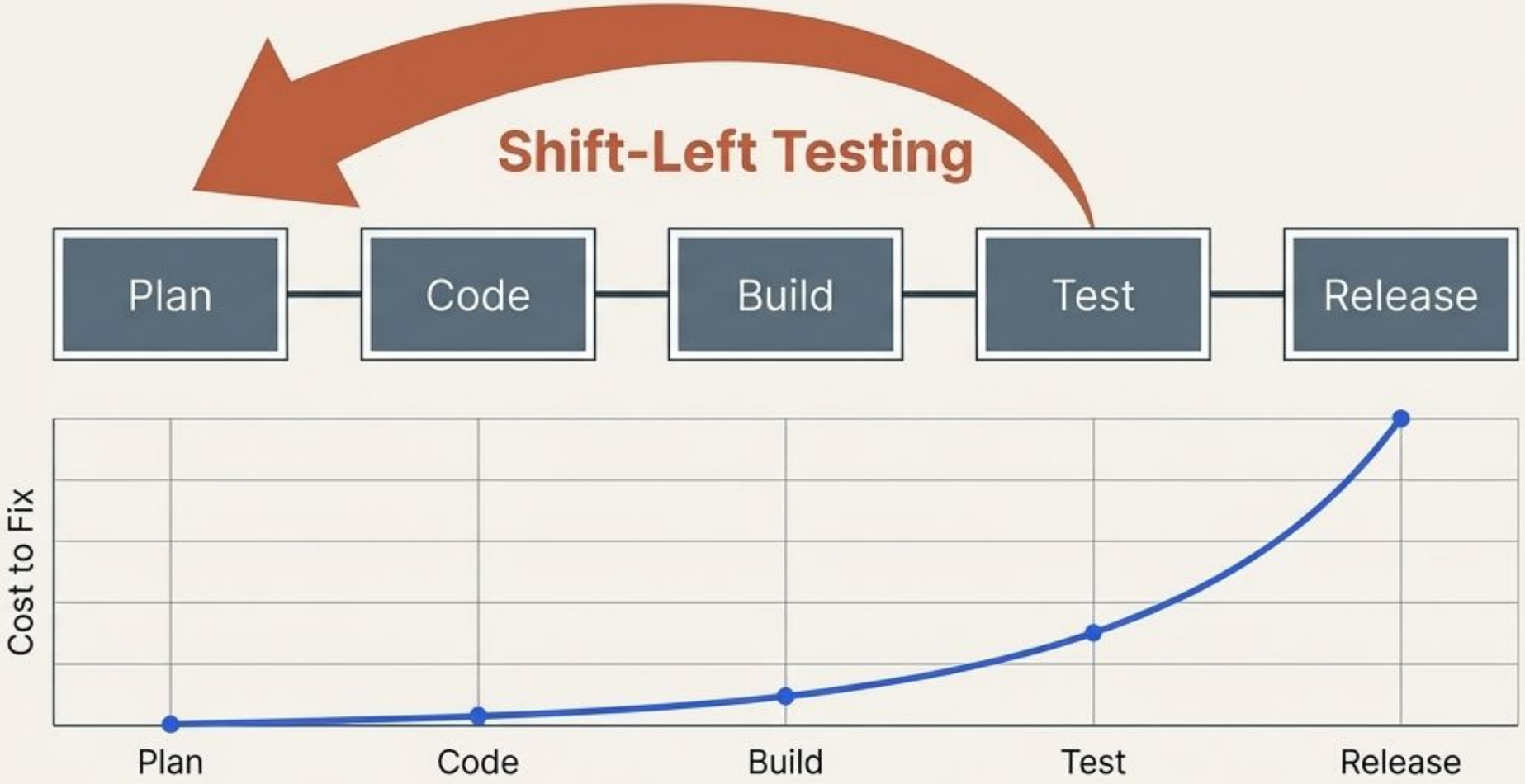
2023 Tesla Autopilot recall over safety defects. This was not just a QA failure, but a breakdown in the fundamental culture of quality during the design stage.



Key Quality Characteristics (ISO/IEC 25010)



Establishing a Culture of Quality



Shared Responsibility

Quality is owned by everyone (Devs, PMs, QA), not just testers.

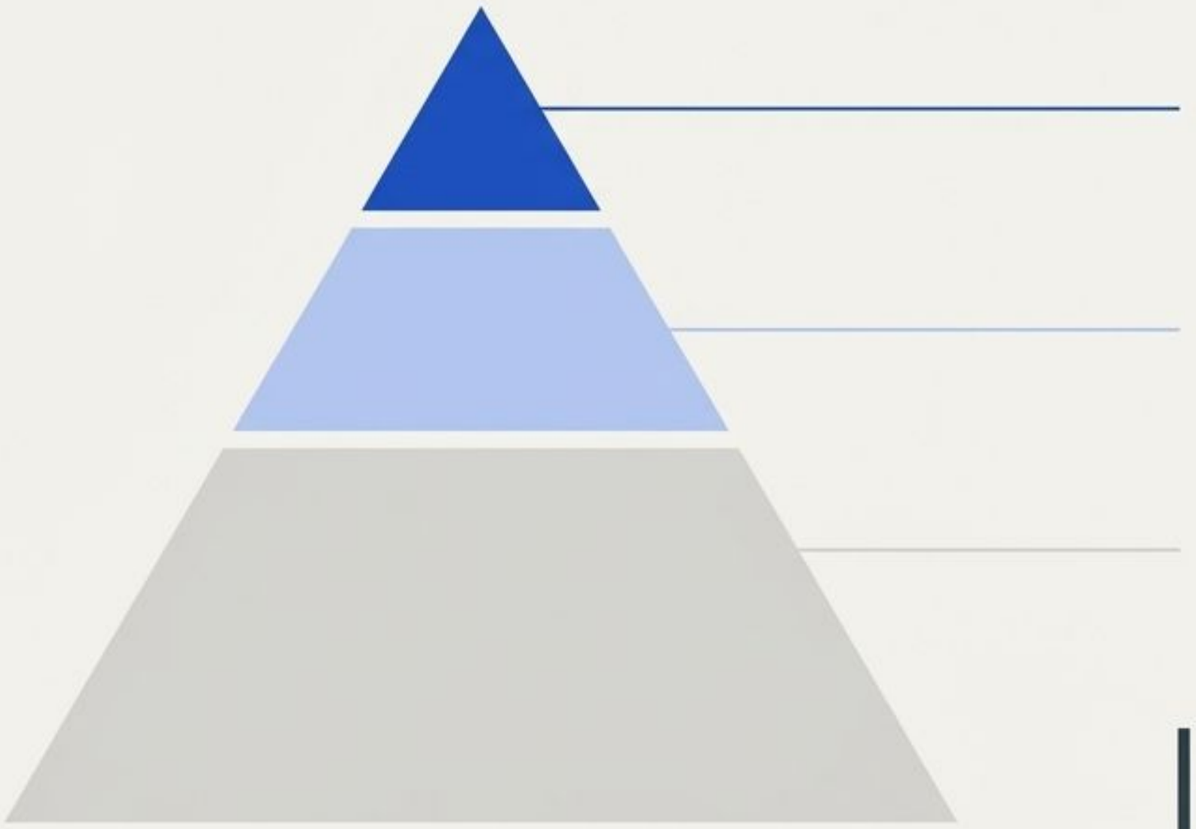
The Shift-Left Strategy

Move testing to the very beginning of the development cycle. Catching issues early is cheapest and safest.

CI/CD Integration

Continuous Integration/Continuous Delivery automates build, test, and deployment for constant, immediate quality checks.

Automated Testing Frameworks



Top (Fewest): End-to-End UI Tests

Tests the entire application flow from the user's perspective.

Middle: Integration/API Tests

Tests how different modules work together.

Base (Most): Unit Tests

Tests individual components/functions.

The Human Element

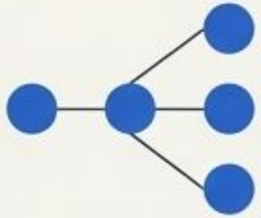
Resilience: Handle crashes without data loss.
Accessibility: Legally & ethically required for users with disabilities.
Exploratory: Real users finding edge-cases scripts miss.

Static Analysis & Quality Metrics

Static Code Analysis
Scanning code for security and performance flaws before it runs.

Failed Deployments
High percentage indicates severe process instability.

Bug Bounce %
Measures customer-reported issues and if the fixes actually work.



Cyclomatic Complexity
Measures the number of paths through a program.
High complexity = exponentially harder to test.

Code Coverage
The exact percentage of code executed by automated tests.

The Gig Economy: Contingent Workers

Definition: Non-permanent freelancers, contractors, or agency staff hired to fill temporary skill gaps.

Agility to scale up/down quickly.

Cost efficiency (no benefits/long-term commitments).

Immediate access to specialized, niche skills.

Follow-the-sun global development speed.

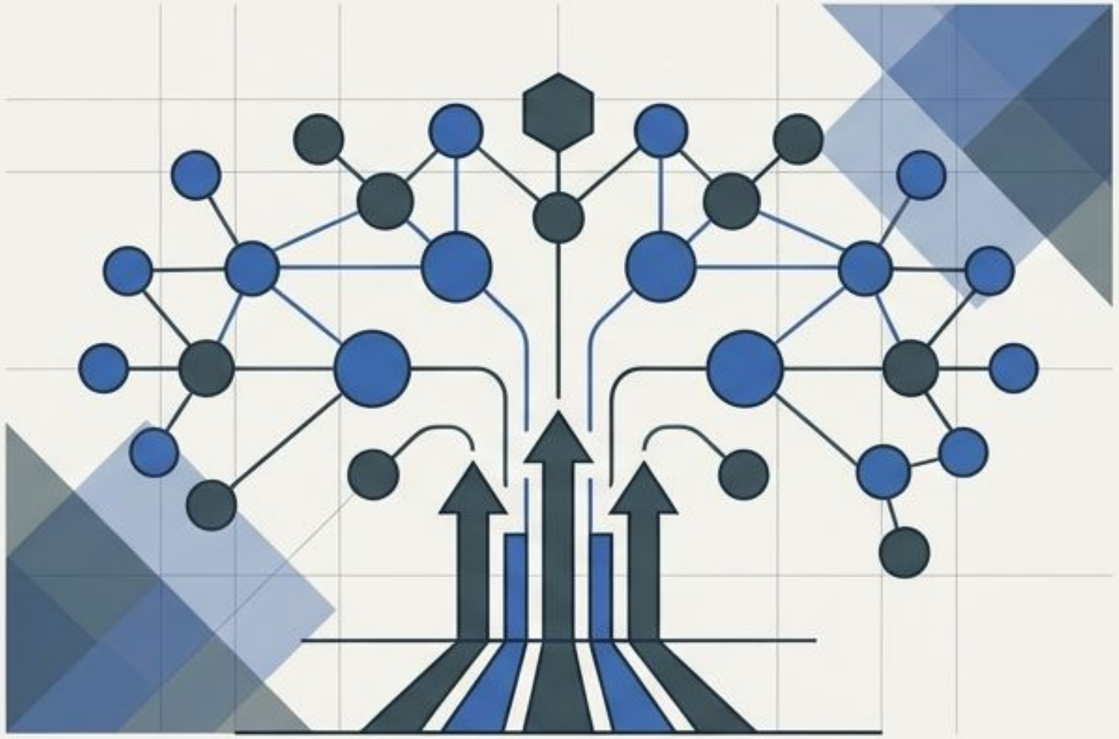
Lower loyalty and long-term commitment.

Friction in cultural fit and communication.

Loss of unique institutional knowledge upon departure.

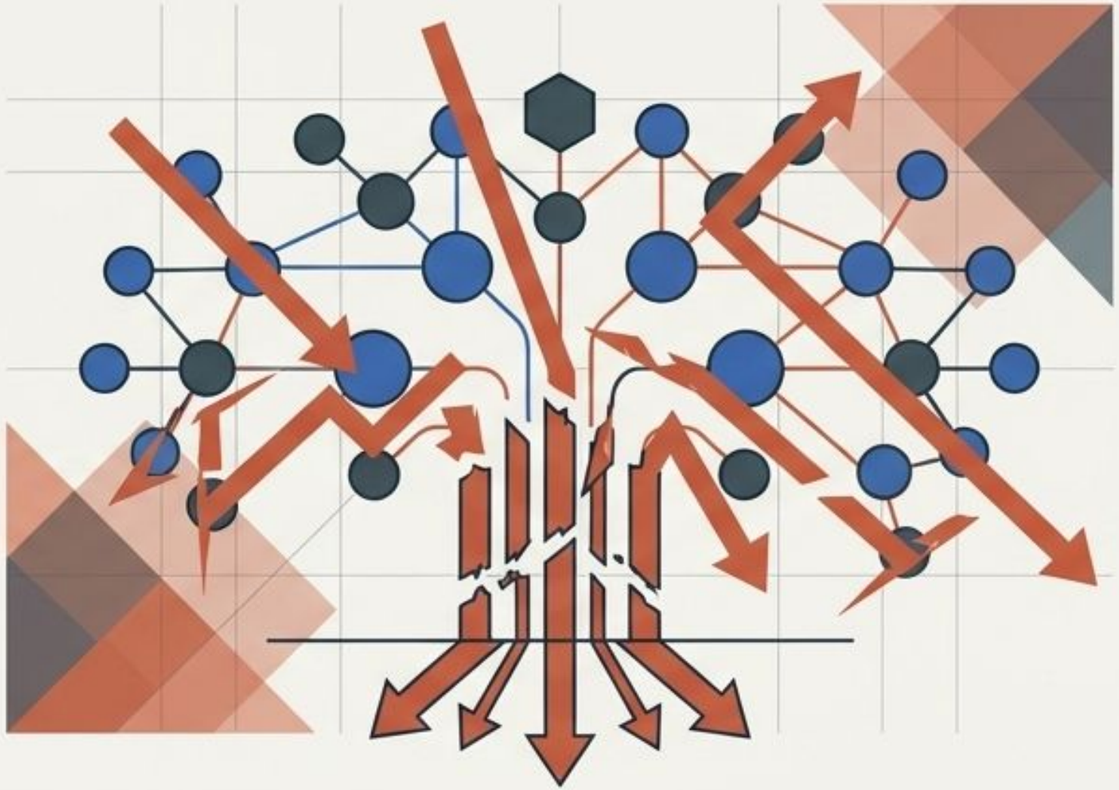
Increased security vulnerabilities if improperly vetted.

The H-1B Dilemma: Intent vs. Reality



The Intended Purpose

- Designed to supplement the US workforce in STEM.
- Provides access to the global best and brightest.
- Fills critical gaps in AI and Cybersecurity to drive innovation.



The Reality of Abuse

- Wage Suppression: Exploited to replace workers with cheaper labor (avg. 36% discount for entry-level).
- Displacement: Domestic IT staff laid off and forced to train foreign replacements.
- High Unemployment: Tech grads face joblessness despite claims of talent shortages.
- Visa Fraud: Outsourcing firms using the program for low-skill, low-wage roles.

Global Outsourcing in IT

2024 Market: $41.6 Billion
(IT accounts for 75%)

The Draw

Cost savings, operational efficiency, access to global expertise, and ability to focus solely on core

The Risks

Loss of direct control, cultural/language barriers impacting quality, and hidden management costs.

Supply Chain Ethics

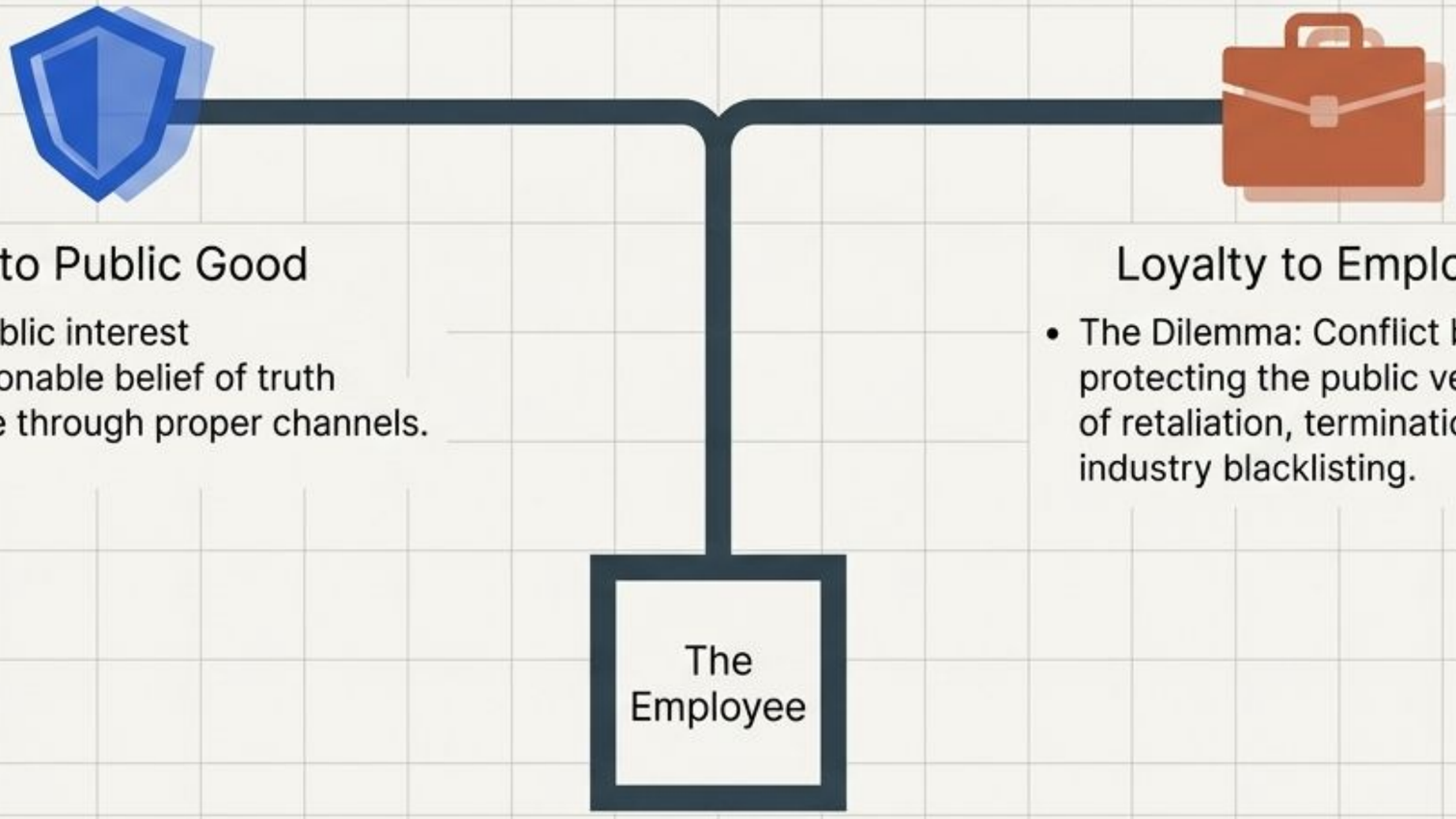
Organizations are ethically responsible for vendor labor practices. Example: Nike's decades of reputational damage over sweatshop labor and poor working conditions overseas.

Whistle-Blowing: The Ultimate Test

Exposing illegal/unethical organizational practices internally (manager/hotline) or externally (media/regulator).

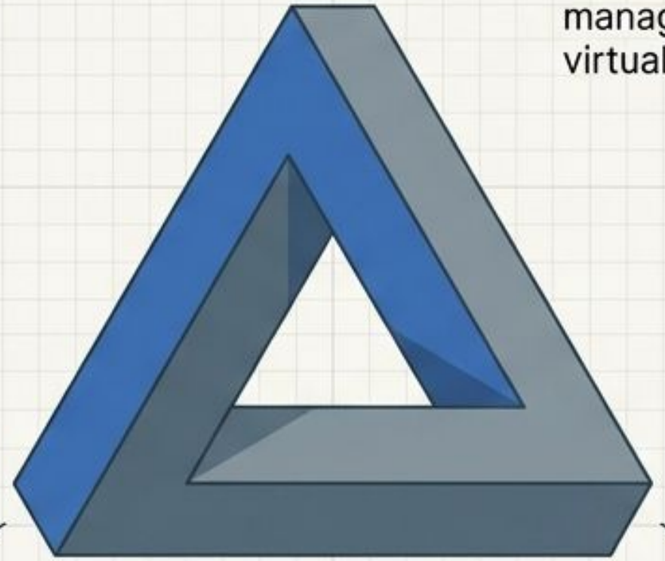
Case Study: Weltec Engineering (Intel Site)

A safety manager reported harness violations. Contract abruptly non-renewed over unrelated voice notes. Highlights the reality of retaliation and corporate cover-ups.



Green Computing Concept

The eco-friendly approach to designing, manufacturing, using, and disposing of IT systems to cut carbon and e-waste.



Reduce

Minimize energy consumption (power management, efficient hardware, virtualization).

Recycle

Safe, certified hardware disposal to recover precious materials and stop landfill e-waste.

Reuse

Extend equipment lifespan (refurbishing, upgrading, donating).

Strategies for Sustainable IT

Organizational Level

- Energy-efficient data centers (e.g., HP Thermal Zone Mapping, Virtualization).
- Device Life Extension (DLE) to refurbish existing tech.
- Procurement policies requiring EPEAT-registered devices.

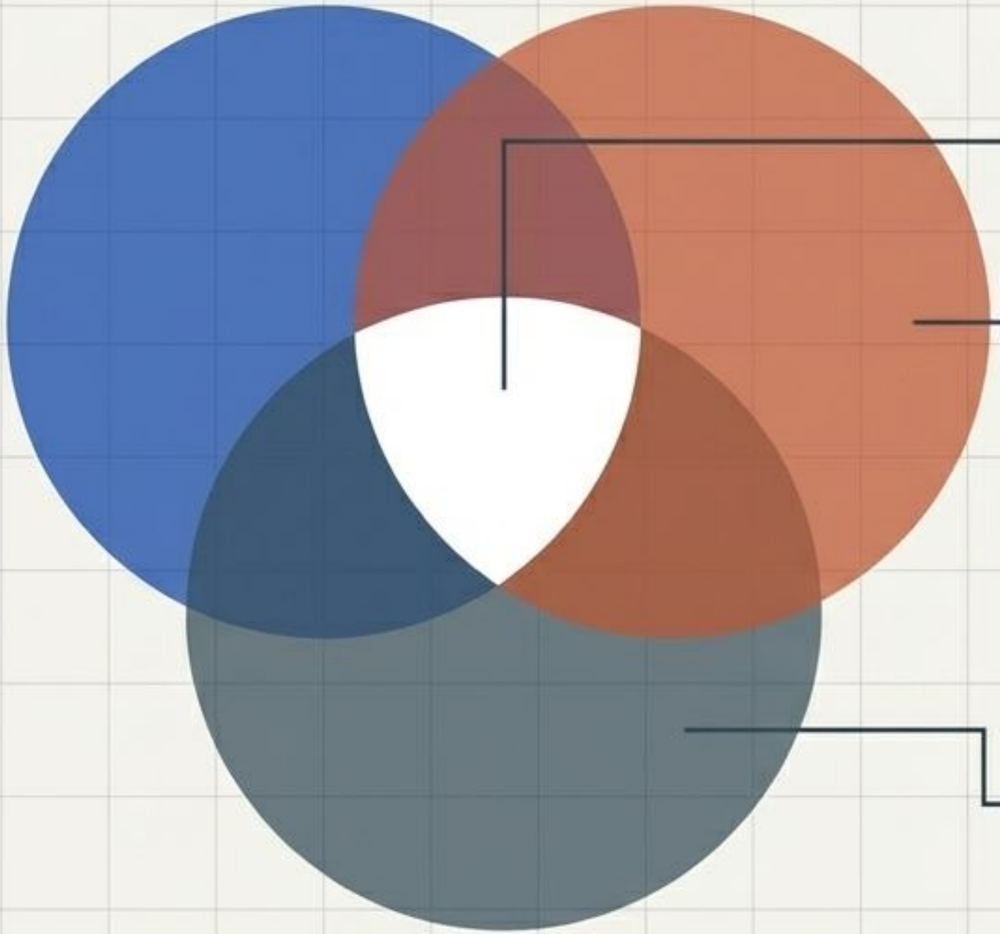
Individual Level

- Enable power-saving settings and nightly shutdowns.
- Double-sided printing only when strictly necessary.
- Strict use of certified e-waste recycling facilities.

The ROI of Green IT (Shared Benefits)

Generates cost savings, ensures environmental compliance, reduces legal risk, and builds a strong ESG-conscious brand reputation (e.g., HP's carbon-neutral computing services).

Synthesis: The Ethical IT Framework



Product Integrity (Code)

Protecting users through Shift-left quality, rigorous testing, and secure architecture.

Workforce Equity (Culture)

Protecting people through fair use of H-1B/Outsourced labor and safeguarding whistle-blowers.

Environmental Duty (Climate)

Protecting the planet through strict adherence to Green Computing limits.

Core Insight: True technological innovation is impossible without alignment across all three ethical pillars. A failure in one is a failure in the system.

Core Exam Takeaways

- **1. Software Quality is a Safety Issue:** Adhering to ISO/IEC 25010 prevents financial and physical harm.
- **2. Shift-Left & Automate:** Moving testing early and utilizing the Test Pyramid is the most ethical, cost-effective strategy.
- **3. Global Labor carries Ethical Risk:** Contingent, H-1B, and Outsourced workers offer agility but risk wage suppression and knowledge loss.
- **4. Whistle-Blowing Requires Protection:** Disclosures must serve the public good, often at immense personal risk of retaliation.
- **5. Green Computing is Mandatory:** The 3 R's (Reduce, Reuse, Recycle) ensure ESG compliance and cost savings.